

The crew plugin: technical reference

The low-level guide, for the people who install, configure and extend crew. A companion overview covers the same ground in plain language. Written 20 September 2026 against crew 0.19.92 on main (commit 621d50dc); the Stop-gate fix (0.19.93) is described as it stands on branch crew-stop-gate-record at b6e8c4bb, in review.

Design constraint. A role earns its place only if it buys one of three things: an isolated context window, a restricted tool set, or genuinely independent eyes. Everything else is a file, a command, or the user. Every custom subagent loads the entire CLAUDE.md hierarchy at startup, so the roster is kept small and the knowledge is kept in files.

1. Layout

Path	Contents	Count at 0.19.92
plugin/crew/commands/	Slash commands, one Markdown file each with YAML frontmatter.	28
plugin/crew/agents/	Role definitions: tools:, model:, description, procedure.	54
plugin/crew/skills/	Procedures loaded on match; each a directory with SKILL.md and references.	20
plugin/crew/hooks/hooks.json and hooks/scripts/	Hook registrations, each event once per shell flavour, and the scripts they run.	20 registrations
plugin/crew/tests/	pytest suites, fixtures, and sabotage.py, which reintroduces known bugs and asserts the suite goes red.	51 modules

plugin/crew/evals/	claude plugin eval cases with free graders; runners in scripts/run-plugin-evals.{sh,ps1}.	5 cases
plugin/crew/CONFIG.md	Every configuration key, its layers and its ratchet rule.	18 sections

2. Roles

The ladder

Tier	Roles	Tools of note	Add when
0	explorer, qa-reviewer	explorer: Read, Grep, Glob only. qa-reviewer: read-only plus Bash.	Always.
1	security, smoke-author, developer	security: no Write or Edit, has Bash. developer: full write, own context, never reviews its own diff.	Security findings reach review; coverage gaps regress; the PM runs work end to end.
2	dba, docs-writer, browser-tester, analyst, planner, infrastructure-architect, scribe, researcher	planner works from an abstracted brief; secondOpinion.sendsCode is false by design.	Per the table in skills/crew-scaling/SKILL.md; /crew:scale reads .crew/metrics.md and recommends.
3	parallel sessions, worktrees, agent teams		Every involved repository has green smoke.

The PM

Standing teammate named crew-pm, reached by SendMessage so it keeps its transcript. Tools: Read, Write, Edit, Bash, Grep, Glob, Agent, Skill. No SendMessage, no ListAgents, so resume-by-id belongs to the caller (commands/pm.md). Authority is report-only | act | autonomous; the stops in crew_state.AUTONOMOUS_STOPS hold at every tier: offboarding a role, deleting

a codemap or diagram, rewriting `.crew/metrics.md`, destroying git history. Dispatch cap 3 per pass. Write scope: `.crew/**`, `TODO.md`, `.work/**`, `docs/diagrams/**`, enforced by the write guard from 0.19.92.

Specialists

`crew_state.SPECIALIST_ROLES` lists 40 roles with no tier, onboarded per repository with `/crew:pm onboard <role>` and justified by the stack (the agent file names the file that proves it). Onboarding a specialist never moves the tier. Groups: languages and frameworks (python-pro, node-developer, php-pro, rust-engineer, react-specialist, angular-architect, dotnet-core-expert, dotnet-framework-4.8-expert, sql-pro, graphql-architect, backend-developer, api-designer); cloud and infrastructure (terraform-engineer, network-engineer, platform-engineer, microservices-architect, legacy-modernizer, database-administrator, git-workflow-manager, workflow-orchestrator, multi-agent-coordinator); Microsoft platform (windows-infra-admin, exchange-online-specialist, sharepoint-developer, power-automate-specialist, powershell-5.1-expert, powershell-7-expert, powershell-security-hardening, ad-security-reviewer); security and compliance (penetration-tester, compliance-auditor, code-reviewer, architect-reviewer, fintech-engineer, payment-integration); review and writing (qa-researcher, kimi-consult, ai-writing-auditor, skill-author, design-bridge).

Model routing

`crew_config.py --models` prints the effective provider, model and family per role, which fallbacks are armed, and whether the self-review guard bars anything. The QA order tries Codex (gpt-6-astra) first, then Copilot (Kimi), then Claude; a reviewer of the same family as the diff's author is struck. Config keys `dev.roles.*`, `qa.roles.*`, `dev.fallback`, `qa.fallback`. The Agent tool's model override applies per dispatch.

3. Commands

Command	Function	Reads / writes
init	Phased, resumable setup; each phase asks before writing.	<code>.crew/config.json</code> , <code>.crew/verify.json</code> , <code>_verify/</code> , CLAUDE.md template.

onboard	Writes the code map; --refresh <subsystem> re-derives one file against the graph.	.crew/codemap/*.md with DERIVED/JUDGEMENT markers and anchor: commits.
ticket, work, split	Scope; implement, review, gate end to end; split an oversized Jira ticket.	.work/tickets/, tracker via MCP.
review	Hostile QA on the diff: Codex, Copilot or a fresh Claude context, first that probes clean; specialist reviewers by path rule.	.work/review/<name>/{diff,prompt,out}.txt; a row in .crew/metrics.md.
verify	Builds the verification map from evidence; --all runs the whole map with no budget; --price (0.19.93) measures and writes seconds.	.crew/verify.json (tracked).
gate	Exports, removes and restores branch protection.	Bitbucket or GitHub API; export file.
promote	Promotes to the next environment; refuses without a .work/PROMOTIONS.md row and, if change.requireForProduction, without a change request.	.work/PROMOTIONS.md.
change	Files a change request in SDP, Jira or .work/changes/; refuses while any of the ten template questions is unanswered.	Tracker via MCP.
emergency	Stands gates down, opens parallel lanes, records what was skipped.	.crew/incidents/.
debug	Systematic debugging: read, reproduce, isolate, prove, then fix. Adapted from superpowers (MIT).	

pm, roster, model, scale, config, upgrade	Manager conversation; role listing; per-role model set and probe; grow or shrink recommendation; config provenance; schema migration (currently to schema 7).	.crew/config.json, ~/.claude/crew/config.json.
plan, survey, docs, reference, diagram, runbook, handoff	Second-opinion design; gap survey; docs touched by a change; API/feature reference with anchors; Mermaid diagrams with %% Anchors;; runbooks; the handoff note.	docs/, docs/diagrams/, .work/HANDOFF.md.
jira-sync, sdp-sync, obsidian-sync	Two-way ticket sync with the local cache.	MCP servers; Obsidian Kanban file.

4. Hooks and guards

Event	Matcher	Script	Behaviour
SessionStart	*	pm-brief, platform sync, context resume	Injects crew state and any handoff note; corrects platform.os/shell.
PreToolUse	Bash, PowerShell	crew_guards.py via shell shims	guards.terraformApply, guards.forcePush, guards.adminMerge (block ask allow); guards.prodDatabase, guards.prodServer (none read full) against production.databases/hosts.
PreToolUse	Write Edit	role_write_guard.py via role-write-guard.{sh,ps1}	guards.roleWrites (off by default; block is the floor). See section 6.
PreCompact	*	handoff writer	Writes .work/HANDOFF.md before summarisation.

Notification	*	pm_pulse.py	Re-briefs the PM with triggers.
Stop	*	verify-gate.{sh,ps1}, context-watch	Runs the matched rules within verify.stopBudgetSeconds (default 60); blocks on failure. See section 7.

Registration rules: each event registered once per flavour with shell:
"powershell" on the Windows entry, because a bare hook command goes to
Git Bash on Windows. Every ratcheted key resolves to the narrower of the
repository and machine-global layers; a corrupt layer resolves to the strict
value; change.requireForProduction ratchets the other way (on wins). A
guard that can block ships off and needs a committed must-block/must-allow
suite, sabotage-tested.

5. Knowledge files

Artefact	Location	Freshness rule
Code map	.crew/codemap/, un-ignored by name	An anchor: behind HEAD means re-check with <code>git diff -name-only <anchor>..HEAD -- <paths></code> ; empty output means current. Comparison is <code>sha[:7] == head[:7]</code> .
Code graph	graphify-out/graph.json and GRAPH_REPORT.md, tracked	Rebuilt by post-commit and post-checkout hooks via <code>graphify update .</code> ; a hand build with <code>--no-viz --code-only</code> writes only one of the pair.
Diagrams	docs/diagrams/*.mmd with % Anchors:	/crew:diagram refresh rewrites only diagrams whose cited files moved.
Metrics	.crew/metrics.md	One row per review; grouped by ticket column for the BLOCK+FIX rate.
Handoff	.work/HANDOFF.md	Injected at session start until finished or deleted.

6. Changes 0.19.90 to 0.19.92

Version	Change	Files and tests
0.19.90	Foreground execution rule in six role files; PM returns the real dispatch id or "not resumable"; idle signal is recency not liveness, two-cycle escalate rule; write-scope sentences aligned with the guard; qa-reviewer's return stays defect-lines-or-CLEAN.	agents/{developer,pm,python-pro,browser-tester,smoke-author,qa-reviewer}.md, commands/pm.md; tests/test_foreground_execution.py (15 tests). Codex: 5, 4+1, 6, CLEAN.
0.19.91	Eval suite: pm-does-not-write-code, qa-reviewer-stays-read-only, developer-defers-unrelated-bug, developer-runs-command-in-foreground, pm-answers-status-mid-pass. Structural graders (ticket as clause subject/object with a whole-clause hedge guard; four-field verdict with a numeric reproduction). Runners: comma xfail list, case-sensitive, exit 127 without CLI, zero cases fails, xfail only for a scored failure from this run with no per-run error, passing xfail fails the gate. Workflow tests the secret via a job-level env flag.	evals/, scripts/run-plugin-evals.{sh,ps1}, .github/workflows/plugin-evals.yml. Live finding: PM case scored 0. Codex: 10; 1+11; 4; 2; 3; 1+1; 1+1.
0.19.92	Role write guard. Policy partition derived from each agent's tools;; test_policy_table_matches_the_agent_files re-derives it. crew_config.layer_state(path) classifies each config layer as absent, ok or corrupt (directory, empty, non-object, guards null/list, dangling symlink); any corrupt layer forces block. Targets resolved component-wise: Windows collapses . . lexically before junctions, POSIX resolves symlinks first; outside-repo and other-drive targets are out of scope and allowed with reason; \\?\ paths refused for scoped roles; stdin read as bytes with BOM stripped; both shims set PYTHONUTF8 and fail closed on launch failure; outer handler cannot raise; foreign namespaces and unknown roles allowed and logged.	hooks/scripts/role_write_guard.py, role-write-guard.{sh,ps1}, crew_config.py, crew_guards.py, CONFIG.md §18; tests/test_role_write_guard.py (114 cases). Codex: 2+4, 7+2, 3+1, 2, 1, CLEAN; PowerShell specialist 2+2.

7. The Stop gate: 0.19.93 in review

Root causes, measured 19 September

- `verify-gate.sh` diffs against `.crew/.verify-verified-at`, written by `record_verified` only when `DEFERRED_COUNT == 0`. `rules[8]` (185 s) against `verify.stopBudgetSeconds 60` deferred every turn; the marker sat at `dedd1150`, 41 commits behind, so 59 files matched 13 rules per Stop. Hot fix applied locally: `marker := gated HEAD`.
- A rule without seconds is never deferrable. Ten of 21 here (PR #197 priced them); 28 of 28 in `TheSelectSource`; 43 of 43 in `SRL`, which also had no marker file and a 499 s Stop.
- In `TheSelectSource`, rules 0–9 run `_verify/smoke.sh` without `--ci`, whose else-branch runs about 18 checks over SSM against the dev web host on every Stop, with `ENV="{ENV: -dev}"` selecting the target from the ambient environment.
- Exit 77 (SKIP under `smoke.sh`) was a hard FAIL under the gate runner.

Components on the branch

Component	Implementation
Per-rule record	<code>verify_record.py</code> : <code>.crew/.verify-gate.record.json</code> , keyed by <code>rule_key()</code> (canonical JSON of paths, run, env, reach, requiresCleanTree; one function for both flavours via the <code>rule-key</code> subcommand). Written atomically; a corrupt or non-dict file poisons the fingerprint so nothing skips; a failed persist withholds both markers; entries whose rule no longer exists are pruned on sync. Sync status starts at 99 and reaches 0 only on a confirmed sync. <code>DEFERRED_COUNT</code> counts acute deferrals only; chronic, reach-excluded, clean-tree and skipped rules stay unverified in the record and are reported every run.
Timings	<code>.crew/.verify-gate.timings.json</code> : <code>max(1, ceil(elapsed))</code> per unpriced command, labelled measured-not-declared; never written to the tracked map by the hook.
<code>--price</code>	<code>verify_price.py</code> , dispatched only on the literal argument; <code>hooks.json</code> passes none or <code>--all</code> . Applies the gate's env pinning and declared env; refuses any rule the Stop path would defer; records 77 as SKIP with no timing.
Reach	<code>verify_record.scan_reach</code> , reject-only: a verb (<code>\bssm\b</code> , <code>\bssh\b</code> , <code>\baws\b</code> , <code>\baz\b</code> , <code>\bgh\b</code> , <code>\bpsql\b</code> , <code>\bmysql\b</code> , <code>\bcurl\b</code> and others, word-bounded) anywhere in the command defers; any script, interpreter with a file argument, inline shell (<code>-c</code> , <code>-Command</code>) or <code>cd</code> defers with <code>"declare reach: local on rules[N]"</code> ; <code>bash -n</code> is parse-only and local; comment-only lines and <code><# #></code> blocks are ignored; <code>python -m X</code> names no file. <code>default</code> and <code>always go</code>

through the same classification. **In progress: split the command on ; && || | & and newlines and classify each segment.** The wrapper-following recursion from earlier rounds is deleted.

Env pinning	ENV, AWS_PROFILE, AWS_DEFAULT_REGION, KUBECONFIG, TF_WORKSPACE unset per rule unless env is declared; identity includes the declared env so two rules with the same text and different env run separately; every Python-to-shell channel writes LF and every bash read site strips CR.
Exit 77, clean tree	SKIP is neither pass nor fail, withholds both markers, and deletes the previous fingerprint before any early exit. requiresCleanTree: true defers to --all with a named reason.
--all	Whole working tree: git ls-files plus staged deletions, rename sources, HEAD diff and untracked.
Triggers	crew_state.read_verify_health: verifyMarkerStale, verifyRulesUnpriced, verifyReachUndeclared; None on a read failure renders as "unknown" and fires. pm_brief.py gains the three findings.
Map and docs	This repository's .crew/verify.json declares reach: local on 18 wrapper rules; test_28 loads the real map and asserts every rule is declared or plainly local. skills/crew-setup/examples/verify-terraform.json carries seconds and reach. CONFIG.md §18 and commands/verify.md describe the record.

Review record

Round	Findings	Notable
1	7 BLOCK, 5 FIX	CRLF from native Python broke env unset in bash; exit 77 advanced markers; --all missed staged deletions; corrupt caches did not poison the fingerprint.
Specialist	1 BLOCK, 1 FIX, 1 NIT	Sync status defaulting to 0 advanced markers with no Python present.
2	2 BLOCK, 6 FIX	Nested wrappers; --price ran wrappers Stop deferred; rule hash omitted env.

3	2 BLOCK, 3 FIX	<code>python -m pytest</code> parsed as a wrapper named <code>pytest</code> , which would have deferred eight rules here every <code>Stop</code> ; caught by the new self-check.
4	4 BLOCK, 1 FIX	Three new ways to hide a wrapper from inspection; design inverted to reject-only.
5	2 BLOCK	Compound commands classified whole; fix in progress.

Operational notes

- Hot fix for an existing repository: `git rev-parse HEAD > .crew/.verify-verified-at`, only if `HEAD` is a verified state. If it is not, this marks unverified commits as verified and the gate will not look at them again.
- After upgrading, declare reach on wrapper rules, run `/crew:verify --price once`, and run `/crew:verify --all` before a PR because a deferred rule is unverified until it runs.
- Open sessions load hooks at start; run `/reload-plugins` to pick up a new version.

8. Known items on the fix list

- `agents/developer.md:178` forbids the commit that every brief asks for; decide once in the role file.
- Developer roles resolve to Claude here although the machine-global config pins Codex; the routing failed silently.
- Every hook's Python writes `stderr` with the OEM code page when piped on Windows; reconfigure to UTF-8 at each entry point.
- `scripts/check-marketplace.py` crashes on a tracked file deleted but not staged.
- `test_verify_gate_lock_window.py` has a load-sensitive timing assertion that fails under concurrent suites.
- The PM cannot name a subagent, so a PM-dispatched developer is not resumable by id.
- Install `scripts/skip plugin/*` on reinstall; CLAUDE.md's `git show/autocrlf` sentence is backwards.

Provenance: counts read from the repository at commit 621d50dc; review figures from `.crew/metrics.md`; Stop-gate details from branch `crew-stop-gate-record` at b6e8c4bb plus the in-progress round-5 fix. Not verified: the Stop-gate fix's effect on live sessions; `drift-detection.sh` was not run for any of this week's merges.